TASK 1: ELECTRICAL DIAGNOSTIC TESTING AND EVIDENCE RECORDING

Task reference: Block_6_Task_1_Electrical_Diagnostic_Check

Task overview

Block
Block 6 - Systems Integration and Advanced Diagnostics

Weeks
Weeks 51-60

Duties covered
1-10

KSBs covered
K12-K19, S8-S15, B1, B2

Estimated duration
60 minutes

Assessment type
Practical / knowledge / documentation

Scenario

A vehicle has an electrical concern requiring structured testing. You must carry out safe electrical checks, record values and explain whether the evidence confirms the fault or requires further diagnosis.

Objectives

- Use a structured electrical testing process.
- Carry out voltage, continuity or resistance checks safely.
- Record test results and compare with expected values.
- Explain findings using evidence rather than assumption.
- Complete diagnostic documentation accurately.

Materials you need at the workstation

- Light vehicle or electrical training rig
- Multimeter and test leads
- Manufacturer wiring or technical information
- Battery support/PPE as required
- Apprentice Assessment Record

Complete all relevant fields. Enter N/A where a field does not apply.

Practical task instructions

1. Confirm the reported electrical concern with the assessor.
2. Prepare the vehicle/rig and select safe test points.
3. Carry out the electrical tests instructed by the assessor.
4. Record results and compare with expected values.
5. Explain whether the evidence confirms the fault and reinstate the workstation.

Vehicle or equipment details

Vehicle make

Mileage / hours

Vehicle model

Engine size / system type

Registration / fleet number

VIN / chassis / equipment ID

Work completed and safe working

Brief description of work carried out and any further recommendations

PPE worn and special precautions taken

Technical data and measurements

Battery voltage

Test point voltage

Voltage drop

Continuity result

Resistance reading

Fuse/relay checked

Relevant DTCs

Confirmed fault/evidence

Special tools including any required calibration

Knowledge questions

1. Why should testing follow a logical sequence?

2. What safety precautions apply when using a multimeter?

3. What can excessive voltage drop indicate?

Knowledge questions continued

4. Why is it important to record actual values rather than just pass/fail?

5. How would you avoid damaging an electrical circuit?

6. When should a wiring diagram or technical data be used?

7. What further action is needed if the test does not confirm the fault?

Assessor observation notes

Record observations, evidence reviewed and any areas requiring further development

TASK 1: BLOCK 6 APPRENTICE ASSESSMENT

Declarations, sign-off and assessment decision

Assessment declarations

Apprentice declaration: I confirm that the work recorded in this assessment document is my own work and that I completed the task under the conditions stated.

Assessor declaration: I confirm that I have observed and/or reviewed the evidence recorded in this document and that the assessment decision is accurate.

Apprentice sign-off

Apprentice name

Date signed

Apprentice signature

Assessor sign-off

Assessor name

Assessor PIN

Date signed

Assessor signature

Sign using your PDF reader's signing tool, or sign by hand after printing.

Assessment decision and validation

Assessment decision - select Pass or Refer; tick resubmission if required

- [ ] Pass
- [ ] Refer
- [ ] Resubmission required

Assessment / evidence reference

Assessor feedback and action points